Tran Khanh Thanh

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Education

VinUniversity — **Hanoi, Vietnam**
Bachelor of Science in Electrical Engineering–GPA: 3.76/4.0 — *Class of 2029*

Honors And Awards

Vietnam AI Innovation Challenge 2026 — **AI for Vietnam & NIC**
Top 10 and FPT Award — *July 2026*

Vietnam AI Contest 2024 — **VLAB Innovation & Boston Global Forum**
First Prize — *Dec 2024*

FIRST Global Challenge 2023–International Olympic Robotics Competition — **FIRST Global**
Gold Medal for Finallist Winning Alliance — *Oct 2023*

Work Experience

STEAM for Vietnam — **April 2025–Now**
Software Engineer & Technical Mentor — *Hanoi, Vietnam*

- **Robotics Tournament Management System:** Built a cross-platform desktop app using Bun, Electrobun, React, Hono, and SQLite with a local LAN HTTP server and real-time SSE synchronization for offline Robotics Tournament operations.
- **Robotics & AI Tutoring:** Mentored 100+ students and high school teams nationwide across STEAM for Vietnam programs and FIRST Tech Challenge (FTC) on Programming, autonomous navigation algorithms, and computer vision pipelines.

Featured Technical & Research Projects

Autonomous Mobile Robot—Indoor SLAM & 3D Reconstruction — **2025–2026**
ROS 2, C++, Python, LiDAR, Depth Camera, Sensor Fusion (EKF), Nav2, SLAM

- Designed and built an autonomous mobile robot from scratch, integrating **multi-sensor fusion** (LiDAR, depth camera, IMU, wheel odometry via EKF) and real-time **SLAM** for robust localization and mapping.
- Engineered **autonomous indoor navigation** with obstacle avoidance, collected synchronized spatial data for **dense 3D reconstruction**, and established the baseline for a **Swarm Robotics** project.

PRISM: A Multi-Dimensional Benchmark for Evaluating LLM Peer Reviewers
Co-first Author – Target: NeurIPS — arXiv:2605.26730 — *prism-benchmark.github.io*

- Developed a multi-dimensional benchmark to evaluate AI-generated peer reviews across 6 quality metrics, comparing 5 LLM baselines against human reviewers over 5 conference datasets.
- Designed an evidence-grounded pipeline for novelty assessment, flaw identification, issue prioritization, and constructiveness using retrieval verification, validated via Pearson, analyses.

Infi—Personal AI for Smarter Stock & Portfolio Research — **2025–2026**
Rust, Tauri v2, React 19, SQLite, Agent Client Protocol (ACP), Bun — *khanhthanhdev.github.io/infi*

- Architected a lightweight local desktop app using Rust and Tauri, implementing the **Agent Client Protocol (ACP)** to orchestrate coding agents (Claude, Codex) over stdio with typed, source-backed analysis blocks.
- Engineered extreme binary size and runtime performance optimizations, deploying custom single-threaded Tokio workers, embedded local SQLite, and sub-second IPC event streaming.

AI Knowledge Cloud (AIKC)— AI Directory Platform — **2025–2026**
Next.js 15, React 19, Bun, PostgreSQL, Prisma, Qdrant, Cloudflare R2, Docker — *aikc.vn*

- Scaled an AI directory of **10,000+ pages** handling **1M+ requests/day** with **100/100 PageSpeed**, using granular ISR/SSG caching, edge CDN routing, and Cloudflare R2 media pipelines.
- Engineered a **hybrid search engine** (PostgreSQL `pg_trgm` + Qdrant) with sub-10ms semantic caching, recommendations, and an automated LLM scraping pipeline to discover and catalog AI tools.

Technical Skills

AI, ML & Data: PyTorch, Transformers, LangChain, Qdrant Vector DB, Scikit-learn, NumPy, Pandas
Languages: Python, C++, TypeScript/JavaScript
Frameworks & Systems: Tauri, Next.js, React, FastAPI, Bun, Docker, Linux, ROS2, Git